

Development of an Intelligent Agent for Optimizing Alarm Deviation in Equipment Health Models Based on Large Language Models

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Abstract—This paper addresses the prevalent pain points in the equipment health management systems of nuclear power units, including difficulties in identifying model alarm deviations (such as missed alarms and false alarms), complex attribution analysis, and low optimization efficiency. It proposes and develops an alarm deviation optimization agent based on a large language model (LLM). With human-machine collaboration as its core concept, this agent integrates cutting-edge technologies including retrieval-augmented generation (RAG), database question-answering, external tool calling, and agent cognitive architecture, and constructs a full-process automated solution covering "deviation detection—root cause analysis—optimization recommendation—closed-loop management." By interfacing with the electronic log database and equipment health management platform of nuclear power plants, the agent automatically compares manual work orders with system alarm information to accurately identify the two types of deviations. On this basis, it innovatively introduces the ReAct reasoning paradigm and counterfactual scenario reconstruction mechanism, and combines with the structural causal model (SCM) to conduct in-depth attribution analysis on multi-dimensional factors such as data pipelines, model logic, and threshold settings. Meanwhile, it leverages RAG technology to enhance the large model's semantic understanding of complex model configurations. Additionally, a multi-source evidence fusion framework is built to achieve self-consistency verification through four dimensions of evidence (numerical, event-based, contextual, and analogical), and the Reflexion mechanism is introduced to support the agent's self-reflection and continuous optimization. Practical applications show that the agent has significantly improved the operational efficiency and accuracy of the platform, providing a replicable technical paradigm for the intelligent upgrade of industrial equipment health management.

Keywords—Large Language Model; Equipment Health Management; Alarm Deviation; Intelligent Agent; Causal Attribution Analysis; Multi-source Evidence Fusion; Human-Machine Collaboration; Closed-loop Optimization.

I. INTRODUCTION

In the context of industrial intelligent transformation, the construction of equipment health management systems has become a core component for enhancing operational reliability and ensuring production safety in critical industrial systems. High-safety, high-complexity industrial systems - represented by nuclear power fields - have universally established equipment health management platforms based on big data analysis and artificial intelligence technologies. These platforms aim to achieve real-time monitoring, fault early warning, and health assessment of critical equipment

components. By deploying numerous state trend monitoring and health degradation models on the platform, they enable refined management of nuclear power plant's vital equipment. However, with the continuous growth in model quantity and complexity, inevitable alarm deviation issues emerge during actual operations^[1], including missed alarms and false alarms. These not only undermine system credibility but may also delay fault response timing, posing potential safety risks.

Traditionally, identifying and optimizing such issues relied on manual comparison between field inspection work orders and platform alarm information, followed by secondary confirmation and root cause analysis. Given that daily work order volumes reach tens of thousands while equipment health platform alarms average only hundreds per day, this process proves time-consuming, labor-intensive, inefficient, and prone to human errors - failing to meet modern industrial systems' requirements for high timeliness and accuracy in operation and maintenance. Therefore, achieving automated and intelligent identification and optimization of equipment health model alarm deviations has become a critical challenge for enhancing platform effectiveness.

In recent years, the rapid advancement of Large Language Model (LLM) technology has provided a novel solution to this challenge. With their powerful semantic understanding, logical reasoning, and multimodal information processing capabilities, LLMs can deeply comprehend unstructured work order texts, parse complex model configuration logic, and combine field operation data for scenario reconstruction and causal inference. Against this backdrop, this study proposes and develops an LLM-based intelligent agent for equipment health model alarm deviation optimization, aiming to construct an intelligent system covering the full process of deviation identification - root cause diagnosis - optimization suggestions - closed-loop management.

Designed with human-machine collaboration as its core philosophy, the agent leverages LLMs as the intelligent engine, integrating Retrieval-Augmented Generation (RAG)^[2], database question answering, and external tool invocation technologies. It achieves efficient and precise identification, analysis, and iterative optimization of measurement point alarms/health degradation model deviations within equipment health management platforms.

The agent first interfaces with nuclear power plant electronic history databases and health management platform monitoring modules to acquire manually reported anomaly

work orders and automatically generated alarm information. Then, utilizing the LLM's semantic understanding capabilities, it automatically identifies real anomaly events described in work orders and matches them with system alarms across dimensions such as time, equipment, and measurement points - thereby detecting two deviation types: missed alarms and false alarms. For each deviation type, the agent further invokes professional analysis tools and domain knowledge bases to delve into technical root causes, including data link issues, model logic defects, and unreasonable threshold settings. Finally, the system generates specific optimization suggestions for model iteration and pushes them to user terminals for confirmation and decision-making, forming a complete closed-loop management process.

This study not only significantly improves the efficiency and accuracy of model deviation identification and optimization but also provides replicable technical paradigms for industrial equipment health management intelligence through its extensible agent framework. This paper will systematically elaborate on the agent's development background, overall architecture design, core functional modules (focusing on missed and false alarm analysis), practical application effects, and future prospects, aiming to offer valuable references for related field research and practice.

II. OVERALL ARCHITECTURE

This paper proposes a large model-based intelligent agent for optimizing alarm deviations in equipment health monitoring models. The system design adheres to multiple construction principles, including human-machine collaboration, standardization and openness, practicality and advancement, flexibility and scalability, and usability. The objective is to establish an intelligent system that is efficient, secure, maintainable, and easily integratable.

The system adopts a hierarchical architectural design, comprising three primary components: Basic Technical Support Layer、Core Functional Module Layer and Intelligent Application Layer, as shown in Figure1^[3].

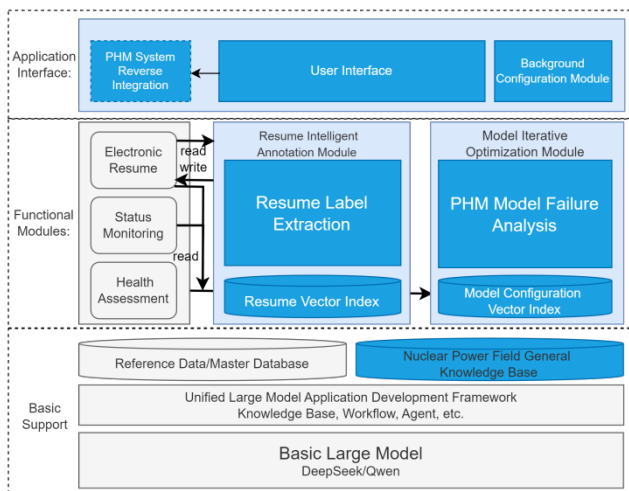


Fig.1. Overall Architecture of the System

The foundational technology support layer serves as the

'power engine' of the entire intelligent agent system, with its core centered on the large model application development technology stack. This project relies on general-purpose foundational large model APIs, such as mainstream high-parameter-scale models like Qwen2.5-72B-Instruct or DeepSeek V3/R1, to ensure the advancement and stability of underlying intelligent capabilities. Building upon this, the system adopts a large model application development framework that integrates core components including general knowledge bases, large model workflows (Workflow), and agents (Agent), providing standardized toolchains for upper-layer application development. As the knowledge input source for large models, this system constructs a specialized vector knowledge base for the nuclear power domain, with its data foundation derived from standardized electronic pedigree reference data/master data specifications (e.g., functional positions, equipment models, failure codes), ensuring agents possess accurate domain knowledge backgrounds when executing tasks.

The core functional layer acts as the 'brain' and 'central processing unit' of the intelligent agent, comprising two submodules: pedigree intelligent annotation and model iteration optimization. The pedigree intelligent annotation module focuses on processing work order information in the electronic pedigree database. By invoking APIs to read work order contents and attachments, it leverages the semantic understanding capabilities of large models to automatically identify critical information such as equipment abnormalities and failure patterns described in work orders, generating preliminary labeling suggestions that are manually confirmed before being written back to the equipment health management system to provide high-quality data input for subsequent deviation analysis. The model iteration optimization module represents the core of this research, responsible for executing critical tasks including deviation detection, root cause analysis, and optimization suggestion generation. This module first reads manually reported anomaly information from the pedigree module and model alarm information from the condition monitoring and health assessment module, then identifies two types of deviations (missed, and false alarms) through comparative analysis. To achieve an in-depth understanding of model alarm logic, the module constructs an independent model vector library for storing parsed operator information, parameter settings, and logical relationships extracted from model configurations in the equipment health management platform, enabling large models to 'understand' the design intent of models. Based on this, the system invokes original data from auxiliary systems like DCS to reconstruct the field environment during anomaly occurrences, combining it with the reasoning capabilities of large models to deduce the root causes of deviations.

The intelligent application layer provides a Web-based interactive interface based on B/S architecture, supporting human-computer collaborative operations between users and agents. The interface includes functional units such as deviation list display, detail analysis, model intent management, and system configuration, where users can review, modify, and confirm analysis results and optimization suggestions automatically generated by large models. To ensure the long-

term effective operation of the system, this research also establishes a continuous operation mechanism for large model applications. Through backend recording of user feedback and large model-generated results, an online issue tracking (trace) system is formed, with accumulated real business data available for subsequent model fine-tuning (Fine-tuning), thereby achieving continuous optimization and iteration of agent capabilities. This overall architecture not only achieves seamless integration and data interaction with the equipment health management system platform but also ensures system flexibility and scalability through modular design and standardized interfaces, laying a solid foundation for future functional expansion and technological upgrades.

III. MISSED ALARM ANALYSIS

In equipment health management systems, a missed alarm refers to a critical deviation where the model fails to respond to equipment abnormalities that have actually occurred. Its essence lies in a semantic disconnection between the model's output and the actual state of the physical world. Traditional manual troubleshooting methods are constrained by fragmented knowledge and broken logical chains, making it difficult to achieve efficient root cause attribution. To address this issue, this paper proposes a missed alarm diagnosis framework based on large-model cognitive reasoning, integrating Counterfactual Reasoning, Structural Causal Modeling (SCM), and Retrieval-Augmented Generation (RAG) technologies. This framework constructs an intelligent agent with the capabilities of "hypothesis-verification-attribution," enabling in-depth penetration and interpretable analysis of the root causes of missed alarms.

A. Missed Alarm Reasoning Engine Based on the ReAct Paradigm

This system builds a ReAct agent architecture centered on Large Language Models (LLMs), modeling missed alarm analysis as a dynamic reasoning process. Driven by "Chain-of-Thought (CoT)," the agent executes multi-step tasks, forming a closed-loop cognitive workflow of "Perception → Reasoning → Action → Reflection"^[4], as shown in Figure 2.

The core workflow is as follows:

1. Perception

The system accesses the nuclear power plant's electronic equipment history database via API to obtain manually reported anomaly work orders.

Using the LLM, it performs semantic parsing on unstructured work order text to extract key entity quadruples: (Equipment, Measurement Point, Abnormal Phenomenon, Time Window).

Example: "Slow rise in X-direction vibration of the non-drive end bearing of the motor for Circulating Water Pump X (February-April 2025)."

2. Hypothesis Generation

Based on the extracted information, the LLM generates preliminary reasoning: "During this time window, the relevant measurement points should have triggered an alarm."

This serves as the starting point for formulating counterfactual hypotheses.

3. External Verification

The agent calls the database interface of the equipment health management system to query alarm records corresponding to the specified time, equipment, and measurement points.

If no relevant alarms or health degradation records are found, the existence of a "missed alarm" is confirmed.

4. Reflection & Planning

The LLM automatically generates a diagnostic task tree: "Is the data not connected? Are features not extracted? Is the threshold not triggered? Or is there a missing logic?"

It then plans the subsequent action path.

This mechanism demonstrates the LLM's capability as a universal task planner. It breaks through the static logic of traditional rule-based systems and enables dynamic responses to ambiguous semantics and complex contextual scenarios^[5].

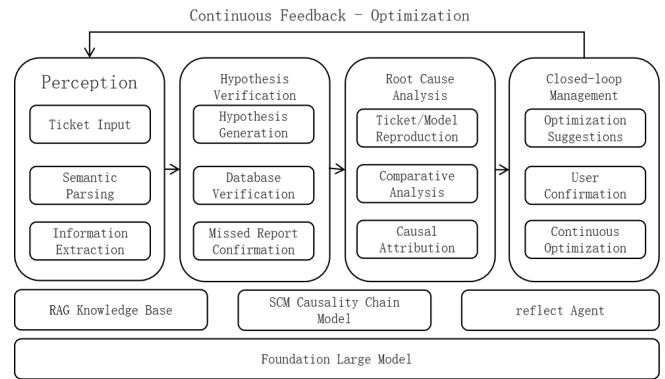


Fig.2. Missed Alarm Diagnosis Framework

B. Counterfactual Scenario Reconstruction and Causal Chain Penetration

To conduct in-depth diagnosis of the root causes of missed alarms, the system implements Counterfactual Scenario Reconstruction. Specifically, under the input of real data, it simulates the ideal response path of "how the model should react if its design were reasonable" and compares this path with the actual output^[6]. The detailed steps are illustrated in Figure 3.

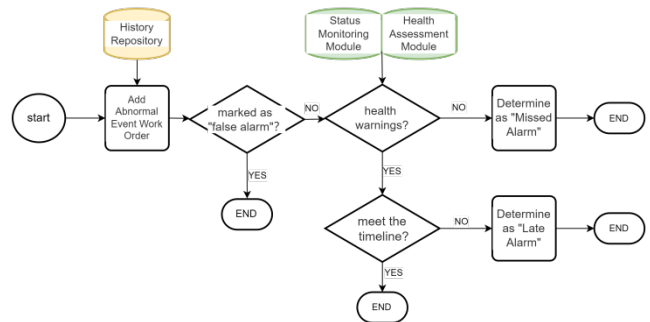


Fig.3. Missed Alarm Analysis Logic

(1)Determine Anomaly Context: This includes defining the precise time range, functional location, equipment number, measurement point name, and anomaly type. This information provides positioning references for subsequent data restoration.

(2)Original Data Restoration: Extract the original measurement point data within the time window specified in the work order from the time-series database of the equipment health management system.

(3) Work Order On-Site Reproduction

Based on the abnormal features described in the work order (such as rapid temperature rise, sudden increase in vibration amplitude, etc.), the system calls the operator API provided by the equipment health management system platform to reproduce the abnormal features specified in the work order (e.g., upward trend of average vibration). This step aims to restore the abnormal phenomena observed by on-site technicians.

(4) Model On-Site Reproduction

Under the same contextual conditions, the system invokes the corresponding monitoring or health assessment model in the equipment health management system, re-executes its complete calculation process, and records the intermediate feature values and final evaluation results.

(5) Deviation Localization

Compare the "work order reproduction results" with the "model reproduction results" to investigate the links where deviations occur layer by layer. For example:Check whether the original measurement points used by the model fully cover the abnormal features involved in the work order;Verify if the model's feature value processing logic (such as filtering, differentiation, aggregation, etc.) is consistent with on-site analysis methods;Assess whether the threshold settings inside the model are too loose, resulting in abnormal signals failing to trigger alarms;Examine if there are flaws in the model's logical judgment chain, leading to the omission of key fault triggering conditions.

To improve analysis efficiency and accuracy, the system pre-summarizes common types of deviation causes, including incomplete component/signal decomposition, lack of fault mode analysis, incorrect threshold settings, missing measurement point monitoring models, data link issues, etc. A knowledge base comparison table is also established to guide the analysis process of the large model.

In addition, to enhance the structure and interpretability of attribution, this paper introduces a causal analysis framework based on Structural Causal Modeling (SCM)^[7], and models the model's alarm logic as the following causal chain:

$$C: D \xrightarrow{f_{\text{link}}} X \xrightarrow{f_{\text{feat}}} F \xrightarrow{f_{\text{thresh}}} A \xrightarrow{f_{\text{eval}}} H \quad (1)$$

Among them:

D: Raw Data (original time-series data of equipment measurement points)

X: Data Link Integrity (state of data transmission, storage, and access links, e.g., whether data is lost during collection)

F: Extracted Features (feature values processed from raw data via algorithms such as filtering, aggregation, or differentiation)

A: Threshold Crossing (result of comparing extracted features with preset alarm thresholds)

H: Health Status (final health assessment output of the equipment, including alarm triggering or normal status indication)

The large model conducts causal attribution analysis by intervening in the input of each step in the causal chain. Typical intervention-based inference scenarios are as follows:

If F is abnormal (i.e., extracted features match the abnormal characteristics described in the work order) but A does not trigger (no threshold crossing is detected), the root cause is inferred to be unreasonable threshold settings (e.g., thresholds are too high to capture abnormal signals).

If X is missing (i.e., raw data cannot be retrieved or is incomplete), the root cause is determined to be data link interruption (e.g., failure of data collection sensors or transmission errors).

If F (Extracted Features) does not include key operators (e.g., failure to apply a vibration trend calculation operator required to identify "slowly rising vibration"), the root cause is identified as model logic defects (incomplete feature engineering design).

This causal analysis method draws on Pearl's Do-Calculus theory. Unlike traditional correlation analysis (which only identifies statistical relationships between variables), Do-Calculus focuses on "interventional effects"—by simulating manual intervention on specific variables (e.g., artificially modifying threshold values to observe changes in A), it eliminates confounding factors and accurately traces the causal direction between variables. This realizes an "intervenable understanding" of the model's internal logic, making the root cause of missed alarms no longer a "black box" inference but a verifiable causal conclusion.

C. RAG-Based Model Intent Understanding and Knowledge Enhancement

To enhance the comprehension of complex model configurations, the system constructs a model vector knowledge base, encoding structured information such as operator logic, parameter settings, and dependency relationships of equipment health models into vectors. The Retrieval-Augmented Generation (RAG) technique provides contextual support for large models.

For example, when analyzing vibration models, the large model can retrieve: "This model supports 20-minute sliding window differentiation but lacks a rate-of-change alarm threshold." Leveraging this knowledge, the model can infer: "The current model relies solely on static thresholds, failing to capture gradual degradation trends," thereby proposing optimization suggestions like "adding rate-of-change alarms."^[8]

This mechanism transforms "black-box models" into "interpretable intentions," significantly improving diagnostic accuracy and explainability. Ultimately, the system generates

concrete iterative optimization recommendations—such as supplementing critical measurement points, adjusting alarm thresholds, or fixing logical flaws—and pushes them to users for confirmation and decision-making, forming a closed-loop management process. Through this automated workflow, the agent notably enhances the efficiency and precision of missed-event identification and analysis, effectively addressing the time-consuming and error-prone drawbacks of traditional manual approaches.

IV. FALSE ALARM ANALYSIS

False alarms occur when equipment health models trigger alerts without real anomalies, leading to the "cry-wolf effect" that severely undermines operators' trust in the system. Traditional methods rely on empirical parameter tuning, lacking systematic diagnostic mechanisms.

A. False Alarm Diagnosis Framework

This study proposes a multi-source evidence fusion and self-consistency verification framework, utilizing large models as "evidence arbitrators" to achieve high-confidence false alarm identification and mechanism-level root cause analysis^[9].

False alarms stem from semantic misinterpretation or oversensitivity to input signals, manifesting as semantic inconsistencies (Semantic Inconsistency) between model outputs and multi-source real-world evidence. To address this, we design an evidence aggregation-based self-consistency verification mechanism, reframing false alarm analysis as a logical judgment of whether "model alerts align with multi-source evidence."^[10]

Upon receiving a new alert, the agent automatically initiates a four-dimensional evidence collection process:

- (1) Numerical evidence: Raw time-series data from the alarm measurement point via DCS;
- (2) Event evidence: Query electronic records for matching abnormal work orders;
- (3) Contextual evidence: Retrieve equipment logs, operational records, and environmental parameters;
- (4) Analogous evidence: Search historical alarm patterns under similar conditions for comparable devices.

As a multi-modal evidence aggregator (Evidence Aggregator), the large model performs semantic alignment and consistency scoring across these four dimensions. For instance:

If no work orders exist, equipment operates smoothly, and similar devices show no alarms → Low consistency → Potential false alarm;

If work orders or operational disturbances occur → High consistency → Valid alert.

Inspired by multi-agent debate (Multi-Agent Debate), this mechanism treats different data sources as distinct "perspectives," with the large model acting as the "referee" for robust decision-making.

B. False Alarm Agent Analysis Process

Using the aforementioned diagnostic framework, the false

alarm analysis agent's workflow is shown in Figure 4. Its trigger conditions include new alarms issued by the equipment health management platform's status warning module or health assessment module. The system first extracts the abnormal context from the alarm information, including the alarm time, functional location, equipment, measurement point, alarm type, and description. Unlike missed event analysis, which starts with manually reported work orders, false alarm analysis begins with system-generated alarms.

The system first determines whether the alarm is indeed a false alarm. The first judgment method is to directly check the alarm's status—if the alarm is actively marked as "false alarm" and closed later, it can be directly confirmed as a false alarm. The second method is to query the nuclear power plant's work order history database. If an abnormal work order matching the alarm time, equipment, and phenomena is found, the alarm is considered valid and not a false alarm. If neither of these two scenarios occurs—meaning the alarm was neither confirmed as false nor was any related work order found—further confirmation through data reconstruction is required.

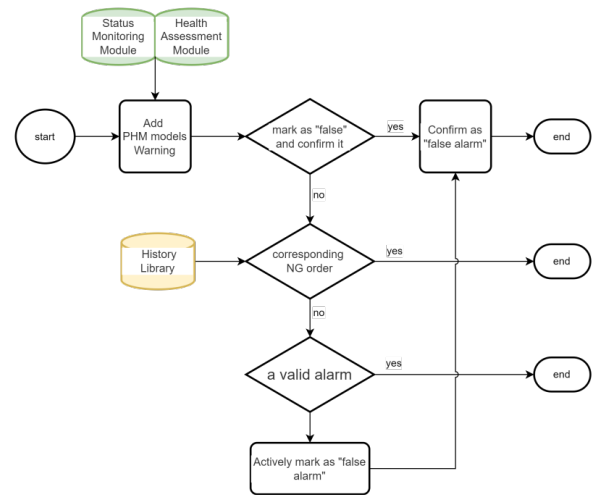


Fig.4. False Alarm Analysis Logic

At this point, the system enters the false alarm deviation analysis process. The first step is abnormal context localization, which involves acquiring all information related to the alarm. The second step is abnormal on-site data acquisition: based on the time and measuring point information in the alarm context, the original time-series data for the corresponding time period is extracted from the Distributed Control System (DCS). The third step is on-site model reproduction—using the acquired original data, the corresponding model in the Equipment Health Management System is re-called to execute its complete calculation process, while recording intermediate feature values and final assessment results to verify whether the alarm trigger conditions are indeed met. The fourth step is result comparison and analysis, which is the core link of false alarm analysis.

By analyzing the results of on-site model reproduction, the system focuses on investigating the following types of causes:

First, whether the features used by the model are reasonable, and whether there are false feature inputs caused by measuring point signal noise, data anomalies, or configuration errors.

Second, whether the processing configuration parameters of the features are appropriate—for example, filter parameters set too low leading to ineffective noise suppression, or differentiation algorithms being overly sensitive to minor fluctuations.

Third, whether the assessment logic for the final results is too aggressive and whether the threshold is set too low, causing fluctuations within the normal range to be judged as abnormal.

In addition, it is also necessary to check whether there are redundancies or errors in the model's logical judgment chain—such as improper priority setting for multiple conditional judgments, or the introduction of irrelevant interference factors. To assist large models in conducting more accurate analysis, the system also constructs a classified knowledge base of false alarm causes, covering common scenarios such as unreasonable threshold settings, model logic defects, data quality issues, and the need for algorithm parameter adjustment.

C. Active Optimization Closed-Loop Based on the Reflexion Mechanism

After completing the cause analysis, the system generates targeted optimization recommendations, such as increasing the alarm threshold, optimizing the feature extraction algorithm, adding a noise filtering link, and adjusting logical judgment conditions. These recommendations, together with the analysis results, are pushed to the user for review and confirmation. Once confirmed by the user, the system can automatically generate a model optimization task and incorporate it into the closed-loop management process^[11].

This system introduces the Reflexion agent architecture to achieve continuous optimization through a "thinking-action-reflection" cycle. After each false alarm analysis, the large model automatically performs self-reflection:

"Was this false alarm caused by an excessively low threshold? If the threshold is increased by 10%, will historical false alarms decrease? Will new missed alarms be introduced?"

By calling the historical data backtesting interface, the large model simulates the performance of the optimization plan on historical data, generates a risk-benefit trade-off matrix, and proposes optimization recommendations within the safety boundary, such as:

Increasing the alarm threshold by 5%;

Adding a 30-second delayed confirmation mechanism;

Introducing sliding window variance filtering

This enables a capability leap of the agent from "passive response" to "active evolution".

It is worth noting that false alarm analysis and missed alarm analysis are not completely independent; they share commonalities in technical methods and knowledge bases. For example, the model intent understanding module provides a common analytical foundation for both, while data restoration and on-site reproduction technologies are universal methods for verifying alarm validity. By integrating false alarm analysis into the agent's automated process, the system can not only promptly identify and correct the model's false alarms but also enhance

the overall robustness and accuracy of the model through continuous optimization iterations, thereby improving the reliability and practicality of the equipment health management platform^[12].

V. APPLICATION VERIFICATION

A. Batch Testing and Quantitative Index Verification

To verify the stability and effectiveness of the intelligent agent in large-scale practical scenarios, batch testing was conducted based on the historical operation and maintenance data of XX Nuclear Power Plant from January 2024 to December 2024. The specific plan and results are as follows:

(1) Source of Test Data and Sample Size

Operation and maintenance data of 3 types of key equipment in the nuclear power plant (circulating water pumps, valves at key positions, and steam generators) were selected, covering 1,200 manually reported abnormal work orders (among which 1,080 work orders were confirmed to have real anomalies after manual review) and 860 alarm records automatically generated by the Equipment Health Management System (among which 192 false alarms were confirmed after manual investigation). A total of 2,060 test samples were included, covering three scenarios: missed alarms, false alarms, and normal alarms.

(2) Evaluation Indicators and Calculation Methods

Missed Alarm Recognition Accuracy: Calculated by the formula "Number of correctly identified missed alarms / Total number of actual missed alarms $\times$ 100%". Among them, actual missed alarms were confirmed by "manually reported abnormal work orders not matched with system alarms", totaling 156 cases; the intelligent agent correctly identified 148 cases, resulting in a missed alarm recognition accuracy of 94.87%.

False Alarm Recognition Accuracy: Calculated by the formula "Number of correctly identified false alarms / Total number of actual false alarms $\times$ 100%". Actual false alarms were confirmed by "system alarms not matched with real abnormal work orders and no anomalies found in on-site verification", totaling 192 cases; the intelligent agent correctly identified 185 cases, resulting in a false alarm recognition accuracy of 96.35%.

Processing Efficiency: The total processing time of a single sample, covering the whole process of "data access - deviation identification - root cause analysis - optimization recommendation generation", was counted and compared with the traditional manual processing method. The average processing time per sample for the intelligent agent was 1.8 minutes, while the average processing time per sample for the traditional manual method (requiring two engineers to collaborate in data verification and root cause analysis) was 45 minutes. The processing efficiency of the intelligent agent was improved by approximately 25 times.

(3) Stability Verification

Dynamic batch testing was conducted continuously for 30 days, with 70 samples randomly selected daily (covering 3

types of equipment). The fluctuation range of missed alarm recognition accuracy was 92.1%-95.3%, the fluctuation range of false alarm recognition accuracy was 94.5%-97.2%, and the fluctuation of processing efficiency did not exceed ± 0.2 minutes per sample. This verifies the stability of the intelligent agent in long-term operation.

The following uses two cases of false alarm and missed alarm to specifically illustrate the working process and application effect of the intelligent application.

B. Verification of Specific False Alarm Case

To verify the capability of the intelligent application in identifying and optimizing false alarm scenarios (where the system triggers alarms incorrectly without actual equipment anomalies), a typical false alarm case of the inlet temperature

of 1# gearbox in XX Power Plant is analyzed as follows:

1) Case Background

The gearbox is a key component of the power transmission system in a power plant. During normal operation, the inlet temperature of the gearbox is monitored by a temperature sensor and stably maintained within the range of 42 – 43 °C. However, on July 1, 2025, an abnormal temperature fluctuation occurred: starting from 03:11:38, the temperature rose sharply, and by 03:12:02, it soared to an extremely high value (183,099,310,080 °C, as shown in Figure 6); after reaching the peak, it quickly dropped back to the normal range. During this process, the gearbox showed no abnormal operating phenomena such as abnormal noise, vibration, or load fluctuation, which belongs to a typical "false alarm scenario caused by non-equipment factors".

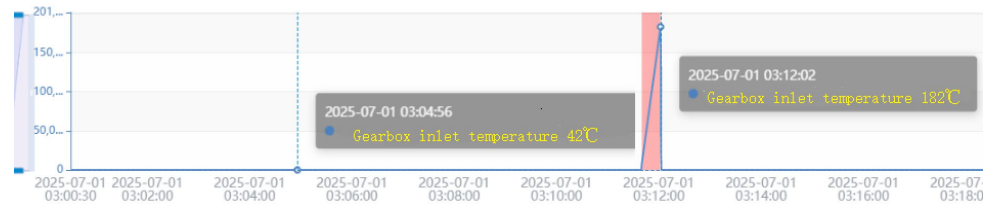


Fig.5. Abnormal Temperature Change Process of Gearbox

2) Diagnostic Process of the Intelligent Agent

When the Equipment Health Management System initially triggered a "high temperature alarm" for the inlet of 1# gearbox, the intelligent application intervened in the analysis through the following steps:

(1) Multi-source data fusion and anomaly judgment

The agent first collects real-time data from associated systems:

① Temperature sensor data: It is confirmed that the temperature abnormally soared from 42 – 43 °C to about 183 billion °C, and then dropped rapidly.

② Vibration/noise monitoring data: The vibration intensity and noise frequency of the gearbox did not show abnormal fluctuations during the temperature rise.

③ Load and process parameter data: During the temperature rise, the load (torque, speed) of the gearbox and the upstream and downstream process parameters (such as lubricating oil pressure, flow rate) remained stable.

(2) Reasoning based on physical constraints

With the help of physical knowledge integrated by the Large Language Model (LLM) (for example, the melting point of common gearbox materials such as alloy steel is below 1600 °C, and the maximum allowable operating temperature of gearbox lubricating oil is usually <150 °C), it is deduced that: the measured temperature of about 183 billion °C far exceeds the physical limit of any gearbox component or medium, and the equipment itself cannot reach this temperature even under normal or even extreme fault conditions. The change process of temperature "rising rapidly to an ultra-high value and then dropping immediately" is inconsistent with the law of "gradual

accumulation/dissipation" of heat in mechanical equipment, but is more in line with the characteristics of abnormal data transmission (such as signal interference, gateway error).

(3) Root cause hypothesis and recommendation generation: Based on the above analysis, the intelligent application determines that the abnormal temperature is not caused by equipment failure, but by abnormal numerical transmission, and generates an optimization recommendation: "Prioritize the inspection of the data transmission loop (including sensors, cables, gateways) of the inlet temperature measuring point of 1# gearbox to eliminate false alarms caused by transmission errors."

3) Root Cause Verification and Effect

After receiving the recommendation, the maintenance personnel carried out targeted inspection on the data transmission loop:

First, verify the temperature sensor: The calibration test shows that the output signal of the sensor is stable at 4 – 20 mA (corresponding to the normal temperature range of 42 – 43 °C), eliminating the problem of sensor drift.

Then, check the gateway device for data transmission: It is found that during the period from 03:11 to 03:12 on July 1, the gateway experienced transient data packet disorder due to software abnormality, causing the temperature value to be incorrectly encoded and transmitted as an "ultra-high value".

After troubleshooting and restarting the gateway, the temperature monitoring data returned to normal (stabilized at 42 – 43 °C), and no similar false alarms occurred subsequently. This verifies the accuracy of the intelligent application's judgment and the effectiveness of the optimization recommendation.

4) Conclusion

This case shows that: The intelligent application can effectively identify false alarms by fusing "multi-source operation data" and "physical knowledge constraints"; it not only avoids "unnecessary equipment shutdown/maintenance caused by false alarms", but also accurately locates "non-equipment root causes (such as data transmission errors)", providing reliable support for the stable operation of the power plant.

C. Verification of a Typical Missed Alarm Case

To comprehensively evaluate the practical effectiveness and reliability of the developed large model-based intelligent agent for alarm deviation optimization of equipment health models, this study selected a typical missed alarm case—abnormal vibration of the non-driving end bearing of the No. X circulating water pump motor in the No. X unit of XX Power Plant—for in-depth application verification^[13].

This case originated from an abnormal work order during actual operation. Through the vibration trend monitoring system, the main control room detected that the average vibration values of the X/Y/Z-axis bearings at the non-driving end of the circulating water pump motor had slowly increased from approximately 1.5 mm/s to 2.4 mm/s since February 2025, after which they stabilized. Although the values did not reach the alarm threshold (4.5 mm/s) or the shutdown threshold (7.1 mm/s), they already indicated an obvious deterioration trend, as shown in Figure 5.

The results of on-site re-measurement were consistent with the monitoring data. Spectrum analysis showed that the vibration was dominated by the rotation frequency, with no other abnormal frequencies. A preliminary analysis suggested the causes might be changes in the equipment support stiffness or the evolution of the bearing condition over operation time. Professionals from various disciplines assessed that the equipment could continue to operate, and a monitoring plan and intervention pre-plan had been formulated.

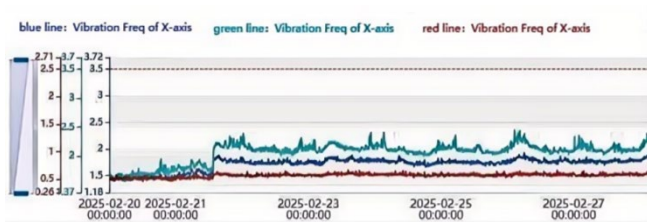


Fig.6. Vibration Deterioration Trend of xx device

However, the Equipment Health Management System platform failed to trigger any measuring point alarm or health degradation notification during the development of this anomaly, resulting in a typical missed alarm scenario. This provided an ideal test case for verifying the intelligent agent.

First, the intelligent agent system automatically accessed and parsed the detailed content of the abnormal work order via the electronic history database interface of the Equipment Health Management System. Leveraging the semantic understanding capability of the large model, it accurately extracted key contextual information, including: the abnormal

equipment (No. X circulating water pump motor), abnormal component (non-driving end bearing), abnormal phenomenon (slowly rising X/Y/Z-axis vibration), abnormal measuring points (X/Y/Z-axis vibration frequency), and abnormal time window (since February 2025).

Subsequently, the system initiated the missed alarm detection process. It queried the alarm records of the equipment and measuring points in question within the aforementioned time window from the condition monitoring and health assessment module of the Equipment Health Management System. After confirming that no relevant alarms or health degradation events had occurred, the system automatically classified the case as a "missed alarm" and generated a deviation analysis task.

Entering the deviation cause analysis phase, the intelligent agent first retrieved the measuring point configuration information from the Equipment Health Management System, obtaining key parameters such as the measuring point codes, source system, and real-value alarm thresholds (High 1 alarm thresholds: 3.5 mm/s, 3.5 mm/s, and 3.0 mm/s respectively) for the X/Y/Z-axis vibration frequencies of the non-driving end bearing of the circulating water pump motor. Simultaneously, the system extracted the original vibration frequency data of these three measuring points from the DCS time-series database starting from February 21, 2025.

Based on this data, the intelligent agent performed two core operations: "work order on-site reproduction" and "model on-site reproduction."

In the work order on-site reproduction: Using the operator API of the Equipment Health Management System, the system processed the extracted original data and calculated the average vibration values from February to April 2025. This successfully reproduced the deterioration trend described in the work order, i.e., "the average value increased from 1.5 mm/s to 2.4 mm/s."

In the model on-site reproduction: The system called the health assessment model for the relevant measuring points in the Equipment Health Management System, used the same original data as input, and re-executed the model's alarm logic judgment process. The analysis results showed that although the measured values (e.g., 1.842 mm/s for the X-axis and 2.213 mm/s for the Y-axis) had increased significantly compared to the baseline values, none exceeded the preset High 1 alarm thresholds (3.5 mm/s). Consequently, the model did not trigger an alarm, which was consistent with the system log records.

Furthermore, the intelligent agent accessed its built model vector database to parse the configuration logic of the health assessment model. It confirmed that the model relied solely on a single criterion—"whether the real value exceeds a fixed threshold"—and was not equipped with a dynamic alarm mechanism based on "rate of change" or "deterioration trend."

Synthesizing the above analysis, the intelligent agent used the large model's reasoning capability to comprehensively determine the root causes of this missed alarm:

The measuring point alarm strategy was overly simplistic. It only set static High 1 alarm thresholds based on absolute values, without configuring thresholds based on the rate of

change. This made it unable to detect slowly evolving deterioration trends, as shown in Table.1.

The early warning logic of the health assessment model was incomplete. Although the model had the capability to calculate the derivative (i.e., acceleration) of vibration frequency over 20 minutes, the internal threshold was set too high or not effectively linked to the health degradation logic. As a result, alarms for deviations in the rate of change also failed to trigger.

Table 1. Alarm Thresholds of the Health Model

Measuring Point Name	Measured Value (mm/s)	Alarm Triggered	Real-Value Alarm Threshold (mm/s)	Rate-of-Change Alarm Threshold
Vibration Frequency of Motor Non-Driving End Bearing (X-axis)	1.84	no	3.5	/
Vibration Frequency of Motor Non-Driving End Bearing (Y-axis)	2.21	no	3.5	/
Vibration Frequency of Motor Non-Driving End Bearing (Z-axis)	0.61	no	3.0	/

Finally, the intelligent agent generated specific iterative optimization recommendations:

At the measuring point alarm configuration level: On the basis of maintaining the existing real-value alarm thresholds, it is recommended to add a "Rate of Change Deviation Alarm Threshold - High 1" to detect rapid or continuous increases in vibration values;

At the health assessment model level: It is recommended to adjust the health degradation threshold for vibration frequency, and correspondingly adjust the health degradation threshold for the 20-minute vibration frequency change rate, enabling it to respond more sensitively to slow deterioration trends.

These recommendations, along with the detailed analysis process and data comparison charts, were pushed to relevant professional engineers for review. The engineers confirmed the accuracy of the analysis results, adopted the optimization recommendations, and subsequently implemented the corresponding model parameter adjustments in the Equipment Health Management System platform.

This application verification not only successfully reproduced the conclusions of manual analysis but also significantly shortened the cycle from problem discovery to optimization recommendation generation through an automated and standardized process. It fully demonstrates the intelligent agent's ability to identify missed alarms, diagnose causes, and generate effective optimization solutions in real industrial scenarios, verifying the rationality and practicality of its design^[14].

VI. CONCLUSION

In summary, this study has successfully developed a large model-based intelligent agent for alarm deviation optimization of equipment health models, effectively addressing core pain points in traditional equipment health management platforms during model operation and maintenance, such as low efficiency, high error susceptibility, and slow response. With

human-machine collaboration as its core concept, the intelligent agent fully leverages the powerful semantic understanding, logical reasoning, and multi-modal information processing capabilities of large language models. Combined with advanced technologies including Retrieval-Augmented Generation (RAG), database question answering, and external tool calling, it constructs a full-process intelligent system integrating deviation detection, cause analysis, optimization recommendation generation, and closed-loop management.

In terms of overall architecture, the system adopts a layered design, achieving a clear separation of basic support, core functions, and application interfaces. This ensures the system's modularity, maintainability, and scalability. In terms of core functions, the intelligent agent enables accurate identification and in-depth analysis of three types of alarm deviations: missed alarms and false alarms. By connecting to the electronic history and condition monitoring modules of the equipment health management system platform, the system can automatically acquire manual work orders and system alarm information, and combine on-site raw data to perform scenario reproduction and causal inference. This allows it to locate technical root causes across multiple dimensions, such as data links, model logic, and threshold settings. After user confirmation, the finally generated optimization recommendations can be incorporated into the closed-loop management process to drive continuous iteration and optimization of the model.

This study not only provides a practical solution for the intelligent upgrading of equipment health management system platforms but also offers valuable experience and technical paradigms for the digital and intelligent transformation of the industrial equipment health management field by constructing a reusable intelligent agent framework. In the future, further exploration can be conducted on the application of large models in more complex scenarios, such as multi-agent collaboration, dynamic threshold optimization, and fault root cause analysis, to continuously deepen the integrated innovation of artificial intelligence and industrial operation and maintenance^[15].

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